Topology: MAT307/M405/MAT803
Problem Set 5 — Solutions to Problems 7–10

Problem 7

Problem. Let X be a compact metric space. Prove that if $f : X \to X$ is an isometry (distance-preserving map), then f is a homeomorphism.

Proof. Let d denote the metric on X, so that $d(f(x), f(y)) = d(x, y)$ for all $x, y \in X$.

Step 1: f is continuous and injective. Given $\varepsilon > 0$, take $\delta = \varepsilon$; then $d(x, y) < \delta$ implies $d(f(x), f(y)) = d(x, y) < \varepsilon$, so f is continuous. If $f(x) = f(y)$, then $d(x, y) = d(f(x), f(y)) = 0$, so $x = y$; hence f is injective.

Step 2: f is surjective. Suppose, for contradiction, that there is $y \in X \setminus f(X)$. Since X is compact and f is continuous, $f(X)$ is compact, hence closed in the metric space X. Therefore there exists $\varepsilon > 0$ such that $B(y, \varepsilon) \cap f(X) = \emptyset$, i.e.

$$d(y, f(x)) \geq \varepsilon \qquad \text{for all } x \in X.$$

Define a sequence $(x_n)_{n \geq 0}$ by $x_n = f^n(y)$ (with $f^0 = \mathrm{id}_X$). For $n > m \geq 0$, since f^m is an isometry,

$$d(x_n, x_m) = d\big(f^m(f^{n-m}(y)),\, f^m(y)\big) = d\big(f^{n-m}(y),\, y\big) = d\big(y, f(f^{n-m-1}(y))\big) \geq \varepsilon,$$

because $n - m \geq 1$. Thus any two distinct terms of the sequence are at distance at least ε, so no subsequence of (x_n) is Cauchy and hence none converges. This contradicts the fact that a compact metric space is sequentially compact. Hence f is surjective.

Step 3: f^{-1} is continuous. By Steps 1 and 2, f is a bijection. For $a, b \in X$,

$$d\big(f^{-1}(a), f^{-1}(b)\big) = d\big(f(f^{-1}(a)), f(f^{-1}(b))\big) = d(a, b),$$

so f^{-1} is also an isometry, hence continuous by Step 1.

Therefore f is a continuous bijection with continuous inverse, i.e. a homeomorphism. □

Problem 8

Problem. Let $\mathbb{R}$ be given the cocountable topology. Is the closed interval $[0, 1]$ a compact subspace of $\mathbb{R}$ under this topology?

Answer. No, $[0, 1]$ is not compact.

Proof. Recall that in the cocountable topology, $U \subseteq \mathbb{R}$ is open if and only if $U = \emptyset$ or $\mathbb{R} \setminus U$ is countable. Let

$$C = \big\{c_n = \tfrac{1}{n} : n \in \mathbb{N}\big\} \subseteq [0, 1],$$

a countably infinite set. Define

$$U_0 = \mathbb{R} \setminus C, \qquad U_n = \mathbb{R} \setminus \{c_k : k \neq n\} \quad (n \in \mathbb{N}).$$

Each of these sets has countable complement, so each is open in $\mathbb{R}$. Note that

$$U_0 \cap C = \emptyset, \qquad U_n \cap C = \{c_n\}.$$

The family covers $[0,1]$**.** If $x \in [0,1] \setminus C$, then $x \in U_0$; if $x = c_n$ for some n, then $x \in U_n$.
Hence $\{U_n \cap [0,1]\}_{n\geq 0}$ is an open cover of $[0,1]$ in the subspace topology.
No finite subcover. Let $F \subseteq \{0\} \cup \mathbb{N}$ be finite. Choose $m \in \mathbb{N}$ with $m \notin F$. Then

$$c_m \notin \bigcup_{n\in F} U_n,$$

since U_0 contains no point of C, and for $n \in F \setminus \{0\}$ the only point of C in U_n is $c_n \neq c_m$. So no finite subfamily covers $[0,1]$.
Thus $[0,1]$ is not compact in the cocountable topology. □

Claim (Remark)**.** *In fact, the same argument shows that a subset of* $\mathbb{R}$ *is compact in the cocountable topology if and only if it is finite.*

Problem 9

Problem. Let $f : X \to Y$ be a continuous map between topological spaces. Show that if X is compact and Y is Hausdorff, then f is a closed map.

We use three standard facts, proved here for completeness.

Lemma (A)**.** *A closed subset of a compact space is compact.*

Proof. Let $C \subseteq X$ be closed, X compact, and let $\{U_\alpha\}$ be an open cover of C by open sets of X. Then $\{U_\alpha\} \cup \{X \setminus C\}$ is an open cover of X, so it has a finite subcover. Discarding $X \setminus C$ (which contains no point of C) leaves a finite subcover of C. □

Lemma (B)**.** *A continuous image of a compact space is compact.*

Proof. Let K be compact, $f : K \to Y$ continuous, and $\{V_\alpha\}$ an open cover of $f(K)$. Then $\{f^{-1}(V_\alpha)\}$ is an open cover of K; pick a finite subcover $f^{-1}(V_{\alpha_1}), \ldots, f^{-1}(V_{\alpha_n})$. Then $V_{\alpha_1}, \ldots, V_{\alpha_n}$ cover $f(K)$. □

Lemma (C)**.** *A compact subset K of a Hausdorff space Y is closed.*

Proof. Let $y \in Y \setminus K$. For each $k \in K$, by the Hausdorff property choose disjoint open sets $U_k \ni k$ and $V_k \ni y$. The sets $\{U_k\}_{k\in K}$ cover K, so there are $k_1, \ldots, k_n$ with $K \subseteq U_{k_1} \cup \cdots \cup U_{k_n}$. Then $V = V_{k_1} \cap \cdots \cap V_{k_n}$ is an open neighbourhood of y disjoint from $U_{k_1} \cup \cdots \cup U_{k_n} \supseteq K$. Hence $Y \setminus K$ is open. □

Solution to Problem 9. Let $C \subseteq X$ be closed. By Lemma A, C is compact. Since f is continuous, $f(C)$ is compact by Lemma B. Since Y is Hausdorff, $f(C)$ is closed in Y by Lemma C. Hence f maps closed sets to closed sets, i.e. f is a closed map. □

Problem 10

Problem. Prove that if Y is compact, then the projection $\pi_X : X \times Y \to X$ is a closed map.

Proof. Let $C \subseteq X \times Y$ be closed. We show that $X \setminus \pi_X(C)$ is open.

Let $x_0 \in X \setminus \pi_X(C)$. Then $(x_0, y) \notin C$ for every $y \in Y$. Since $X \times Y \setminus C$ is open, for each $y \in Y$ there is a basic open set $U_y \times V_y$ with

$$(x_0, y) \in U_y \times V_y \subseteq (X \times Y) \setminus C,$$

where $U_y \subseteq X$ and $V_y \subseteq Y$ are open. The sets $\{V_y\}_{y \in Y}$ form an open cover of the compact space Y, so there exist $y_1, \ldots, y_n \in Y$ with

$$Y = V_{y_1} \cup \cdots \cup V_{y_n}.$$

Put $U = U_{y_1} \cap \cdots \cap U_{y_n}$ (with $U = X$ if $Y = \emptyset$). Then U is an open neighbourhood of x_0.

Claim. $U \cap \pi_X(C) = \emptyset$.

Suppose instead that $x \in U \cap \pi_X(C)$. Then $(x, y) \in C$ for some $y \in Y$. Choose i with $y \in V_{y_i}$. Since $x \in U \subseteq U_{y_i}$, we get

$$(x, y) \in U_{y_i} \times V_{y_i} \subseteq (X \times Y) \setminus C,$$

contradicting $(x, y) \in C$. This proves the claim.

Thus every point x_0 of $X \setminus \pi_X(C)$ has an open neighbourhood $U \subseteq X \setminus \pi_X(C)$, so $X \setminus \pi_X(C)$ is open and $\pi_X(C)$ is closed. Hence π_X is a closed map. $\square$